

Two pager on Panorama Localisation

2nd September 2026

The Challenge

Challenge name/tagline: Am I at a unique place? *Where in the world could here be?*

Challenge one-liner: Identify plausible locations of a given 360° panorama using interaction with AI agents.

Introduction

Panorama images offer a rich representation of the surroundings of any point on land since they, to a fair extent, remove the subjectivity of the picture framing by the photographer. Thus, a third person not involved in the image capture should be more capable of locating a 360° panorama than of a photo featuring a narrower field of view. The goal of the challenge is to find *one or multiple plausible locations*.

All photographs are from real places, taken at near ground level (no more than 5m above ground) without human image editing.

The Team will be handed a set of panoramas to prepare the challenge. Some of them will be from easy to spot sites, others from less known places. The preparation images will not be localised and they will follow the same specifications of the final challenge images. The number of preparation images should be similar to the size of the final challenge set of panoramas.

The Team may prepare the software or AI agents or whatever means it finds useful for the final challenge on Sunday where a fresh set of panoramas will be handed. This final scoring run is time limited.

The ideal process is that humans are so efficient at teaching the AI agent to solve the task that “it” can do the final run on its own. Otherwise, humans and the programmed AI agent will collaborate during the final run to get to plausible locations. In case an answer to a given panorama is really close to the actual site, there is a bonus. There is also a “Special One” panorama with a premium scoring.

Multiple plausible locations are welcome, with a modest “cost” related to decreasing likelihood. However, if the AI agent trusts its findings, “it” can repeat one location along the list to improve its score.

The core of the challenge

The goal is to return a list of locations (in lat,lng coordinates) sorted by decreasing likelihood of a dated (YYYYMMDDhhmmss), equirectangular 360° panorama with 6080 x 3040 pixels in JPG format. The panoramas come mainly from Portugal but also from Spain and France.

The Team will be handed a list of N^1 panoramas. Each panorama will be labelled as “Urban” or “non-Urban”. This label serves to select the appropriate distance function, θ_U or θ_{NU} . The final scoring run will take M^2 minutes during which all interactions with the AI agent should be recorded and handed to the jury. The Team can use any tools during the scoring run on other equipment. However, it cannot feed data other than text, figures and multiple choice “clicks” into the AI agent.

The scoring rules are as follows.

1. The list is limited to L^3 entries; each entry contains only one (lat,lng) pair.
2. The geodesic distance between any candidate answer to the correct location is measured in meters using GIS.tools [1]. All distances below 10m will be considered exact: $d=0$.

¹ $N=10?$

² $M=60?$

³ $L=10?$

3. The location at the top of the list is worth a base score of 1000 points that can be modified by multiplicative correction factors applied as stated next.
4. The distance to the goal is a corrective factor θ_U or θ_{NU} , for urban and non-urban panoramas respectively, computed according to the formulas where d is the distance in meters:

$$\theta_U = \frac{2}{1 + \frac{d}{100}} \quad \theta_{NU} = \frac{2}{1 + \frac{d}{1000}}$$

5. Descending each rank in the result list will represent a 0.95 correction factor. Thus, the answer at the R^{th} rank starts at $1000 \times 0.95^{(R-1)}$.
6. The Team can write cues to the AI agents through prompts or other methods up to 1000 characters per cue. Direct or indirect references to locations or features are excluded⁴. For each cue the Team gives to the AI agent, it will represent a 0.8 correction factor.
7. The Team can answer multiple-choice questions proposed by the AI agent up to 4 options per question. These can be about features, about “geographic zones”, etc.. The only restriction is that they are generated by the AI agent and not “suggested” by the Team during the final run. Each multiple question asked by the AI agent will represent a 0.95 correction factor, regardless of the correctness of the answer.
8. In case the AI agent returns a qualitative localization – for instance “Praça do Comércio, Lisboa” or “Município de Beja” – the Team may define a geographic coordinate within the area with a correction factor of 0.5.
9. The correction factors for cues given according to #6, #7 and #8 are applicable to all subsequent candidate answers for that panorama.
10. All panoramas in the final scoring deck have the same weight/relevance but for the “Special One”, which is identified and bears a **3x** weight.
11. The score for each panorama will be the sum of the scores for each location.
12. The final score of the Team will be the sum of the scores for each panorama. In case of tie, the Team which finishes in the shortest time wins.

Nota bene: differences and similarities with popular geo-referenced panorama games

While this challenge resembles GeoGuessr [2], WorldGuessr [3] or TimeGuessr [4] games or alike, some differences should be highlighted, especially for the benefit of regular players:

1. While these games feature panoramas from all over the world, in our case the geographic scope is narrower; hence the ambiguity or confusion between candidate locations should be higher.
2. While many panoramas are captured next to a street or road some of them are in open fields, so don’t restrict your search to roadside image databases.
3. The mention of street or road names is rare in these panoramas, removing a powerful cue.

For the curious mind, the motivation for the challenge is AI related and explained in the companion document [5] that is not relevant to solve the challenge.

References

- [1] <https://gis.tools/distance-calculator>. This site uses the Haversine formula, see https://en.wikipedia.org/wiki/Haversine_formula
- [2] <https://www.geoguessr.com/>

⁴ Example of unacceptable indirect reference: “the most famous sport here is big wave surfing’ referring to Nazaré.

[3] <https://www.worldguessr.com/>

[4] <https://timeguessr.com/>

[5] Praying for Portugal – How the JunctionX challenge began

Future potential and benefits

This is a game, thus the immediate benefit is to have fun while playing the game.

The second level benefit is to foster interactive applications with AI agents where questions and answers come from both sides and humans learn and teach AI agents how each “side” observes one image looking for relevant elements.

The third and ultimate benefit is to identify which features are believed to be singular of landscapes and regions, and yet they resemble features found elsewhere; a more thorough research should highlight the minimal set of features that identify a single region beyond ambiguity.

After the challenge

The *ultimate purpose* of the challenge is to return the AI-assisted deduction process (could we call it *reasoning*?) that led to a set of possible locations ranked by likelihood and obtained from the cooperation of humans and AI agents.

One of the expected outcomes is to find pairs or multi-sets of places that are far apart but yet are quite similar in their landscape or views and probably share many environmental conditions, which are one of the root causes of landscape, the other being humans.

The challenge will be a full success if players conclude that regional affinities are stronger than differences and what fits one environment could also suit other environments and, finally, if one enjoys living *here*, one should also enjoy living *there*.